

Saurabh Kumar Singh

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Right to work in the UK: No employer sponsorship required

ORCID: [0000-0002-2760-9671](https://orcid.org/0000-0002-2760-9671) | Portfolio: sksaurabh34.github.io/my-website)

Professional Summary

Experimental physicist and instrumentation developer with experience in atomic and molecular beam experiments, supersonic source development, precision laser diagnostics, and ultra-high-vacuum instrumentation. Skilled in the design, modelling, construction, and characterisation of compact high-intensity beam sources, interferometric laser diagnostic tools, and custom detection systems for state-resolved beam and gas-surface experiments. Experienced in magnetic manipulation of beams, simulation-assisted instrument development using CAD and MolFlow+, and scientific programming in Python and MATLAB for analysis, modelling, automation, and quantitative performance evaluation.

Core Expertise

- **Atomic and Molecular Beams:** Supersonic beam generation, compact atomic helium beam sources, molecular beam diagnostics, beam collimation, high-intensity low-divergence source development
- **Beam Manipulation and State Control:** Magnetic-field manipulation of He-3 and H₂ beams, polarisation control, alignment and orientation studies
- **Laser Diagnostics and Precision Optics:** High-precision wavemeter development, transfer interferometer concepts, laser frequency stabilisation, interferometry, optical alignment
- **Modelling and Scientific Computing:** Python, MATLAB, SIMION, MolFlow+, CAD-based design, conductance calculations, simulation-assisted design, quantitative data analysis
- **Experimental Instrumentation:** Detector integration, TOF-MS, time-of-flight measurements, velocity map imaging, UHV systems, beamline design, calibration and performance validation

Professional Experience

Postdoctoral Research Assistant

Oct 2024 – Present

Surface Dynamics Lab, Swansea University, Wales, UK

- Developing atomic and molecular beam instrumentation for surface scattering studies.
- Designing and optimising a compact high-flux atomic/molecular beam source using CAD-based design, conductance modelling, and MolFlow+ simulations to guide source geometry and vacuum performance.
- Working on magnetic manipulation of He-3 and H₂ beams for polarisation control, alignment, and orientation surface scattering studies on metal surfaces under UHV environment.
- Designing, installing, and calibrating diagnostic systems for beam characterisation and gas-surface interaction studies under ultra-high vacuum conditions.
- Using Python and MATLAB for analysis, quantitative comparison of experimental performance, and support of instrument optimisation.

Ph.D. Researcher

2018 – 2024

Tata Institute of Fundamental Research (TIFR), Hyderabad, India

- Designed and built molecular and atomic supersonic beam experiments for state-resolved surface scattering studies.

- Developed a compact atomic helium beam source with high intensity and low divergence for precision beam experiments.
- Built and calibrated precision laser diagnostic tools, including a high-accuracy wavemeter based on crossed-fringe Fizeau interferometry.
- Designed and optimised TOF-MS and velocity map imaging systems for beam and surface-scattering diagnostics using SIMION and experimental validation.
- Developed custom optical, ion-detection, and vacuum-based instrumentation, combining numerical modelling, calibration, and performance analysis to guide design decisions.

Selected Projects

Compact Atomic / Molecular Beam Source Development

- Designed and developed a compact beam source for atomic and molecular beam generation, with optimisation guided by CAD design, vacuum conductance modelling, and MolFlow+ simulations.
- Achieved a high-intensity, low-divergence beam suitable for precision beam experiments.
- Identified the potential of the same compact source architecture for use as a deposition source; detailed characterisation is ongoing, including time-of-flight measurements.
- The low beam divergence also makes the source promising for Doppler-reduced or near-Doppler-free spectroscopy geometries, with potential utility in efficient laser frequency stabilisation.

Magnetic Manipulation of He-3 and H₂ Beams

- Worked on magnetic-field-based manipulation of He-3 and H₂ beams for polarisation control and state-selective beam studies.
- Contributed to alignment and orientation control experiments for precision scattering measurements.

Interferometric Laser Diagnostics

- Developed high-precision interferometric diagnostics for laser wavelength and frequency characterisation.
- Designed and implemented optical alignment and calibration strategies for precision laser diagnostics relevant to atomic and molecular beam experiments.

TOF-MS and Velocity Map Imaging Systems

- Designed and built a TOF mass spectrometer based on McLaren space-focusing geometry and developed velocity map imaging assemblies for state-resolved experiments.
- Optimised ion optics in SIMION and validated instrument performance experimentally.

Education

Ph.D. in Chemical Science	2018 – 2024
Tata Institute of Fundamental Research (TIFR), Hyderabad, India	
M.Sc. in Physical Chemistry	2016 – 2018
Banaras Hindu University, India	
B.Sc. (Hons.) Chemistry	2012 – 2015
Banaras Hindu University, India	

Patent

Saurabh Kumar Singh, Avinash Kumar, and Pranav R. Shirhatti

A multistage thermal drift compensated cross-fringed Fizeau interferometer wavemeter and method thereof.

Indian Patent Application No. 202321048766, filed 20-Jul-2023.

Selected Publications

- Saurabh Kumar Singh, Avinash Kumar, and Pranav R. Shirhatti, *A simple and low-cost setup for part per billion level frequency stabilization and characterization of red He–Ne laser*, HardwareX, 2023.
- Geetika Bhardwaj, Saurabh Kumar Singh, and Pranav R. Shirhatti, *A compact and highly collimated atomic/molecular beam source*, Review of Scientific Instruments, 2023.
- Additional publications: see [ORCID](#) and [Research portfolio](#)